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Newton's Laws of Motion: Reading Guide

4.1 Development of Force Concept

Definitions

- dynamics
- force
- free-body diagram

Concept Check

A force is which of these?

- ☐ a vector ☐ a scalar ☐ a quasar ☐ a quaternion

4.2 Newton's First Law of Motion: Inertia

Newton's First Law (NFL)

A body at rest _____ at rest, or, if in motion, _____ at a _____ unless acted on by a _____.

Thinking about the First Law

According to the book, what is the important verb in Newton's First Law, and why is it important?

Concept Check

A trick question! A school bus comes to a sudden stop, and all of the backpacks on the floor start to slide forward. What force causes them to do that?

- ☐ the force of the brakes ☐ friction ☐ the force of inertia ☐ there is no force that causes that!

Explain your answer:

Definitions

- inertia
- mass

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4.3 Newton's Second Law of Motion

Newton's Second Law (NSL)

The acceleration of an object is

- directly proportional to and in the same _____ as the _____ acting on it.
- inversely proportional to its _____

Equations:

Definitions

- inversely proportional
- directly proportional
- weight

Weight and Mass

Summarize what weight is and how it is different than mass.

Equation for weight:

Concept Check

An astronaut on the moon has...

- ☐ less mass than on earth ☐ less weight than on earth ☐ less mass and less weight than on earth

Explain your answer:

4.4 Newton's Third Law of Motion: Symmetry in Forces

Newton's Third Law (NTL)

Whenever one body exerts a force on a second body, the first body experiences a force that is _____ in magnitude and _____ in direction to the force that it exerts.

Equation:

Shortened form: Every _____ has an _____ and _____ reaction.

Examples

Write down three examples of Newton's Third Law in action from the book.

Concept Check

A hammer hits a nail. If the action force is the hammer pushes the nail down, what is the reaction force?

- ☐ The nail moves down
- ☐ The wood pushes the nail up
- ☐ The hammer pushes the nail up
- ☐ The gravity of the hammer pulls it down
- ☐ The nail pushes the hammer up

An airplane experiences a force of gravity in which the earth pulls the plane down. What is the reaction force to this?

- ☐ The airplane's wings push the plane up
- ☐ The airplane pulls the earth up
- ☐ The airplane's wings push the air down
- ☐ The airplane eventually comes down.

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